EVAL 240W-60V with L6790

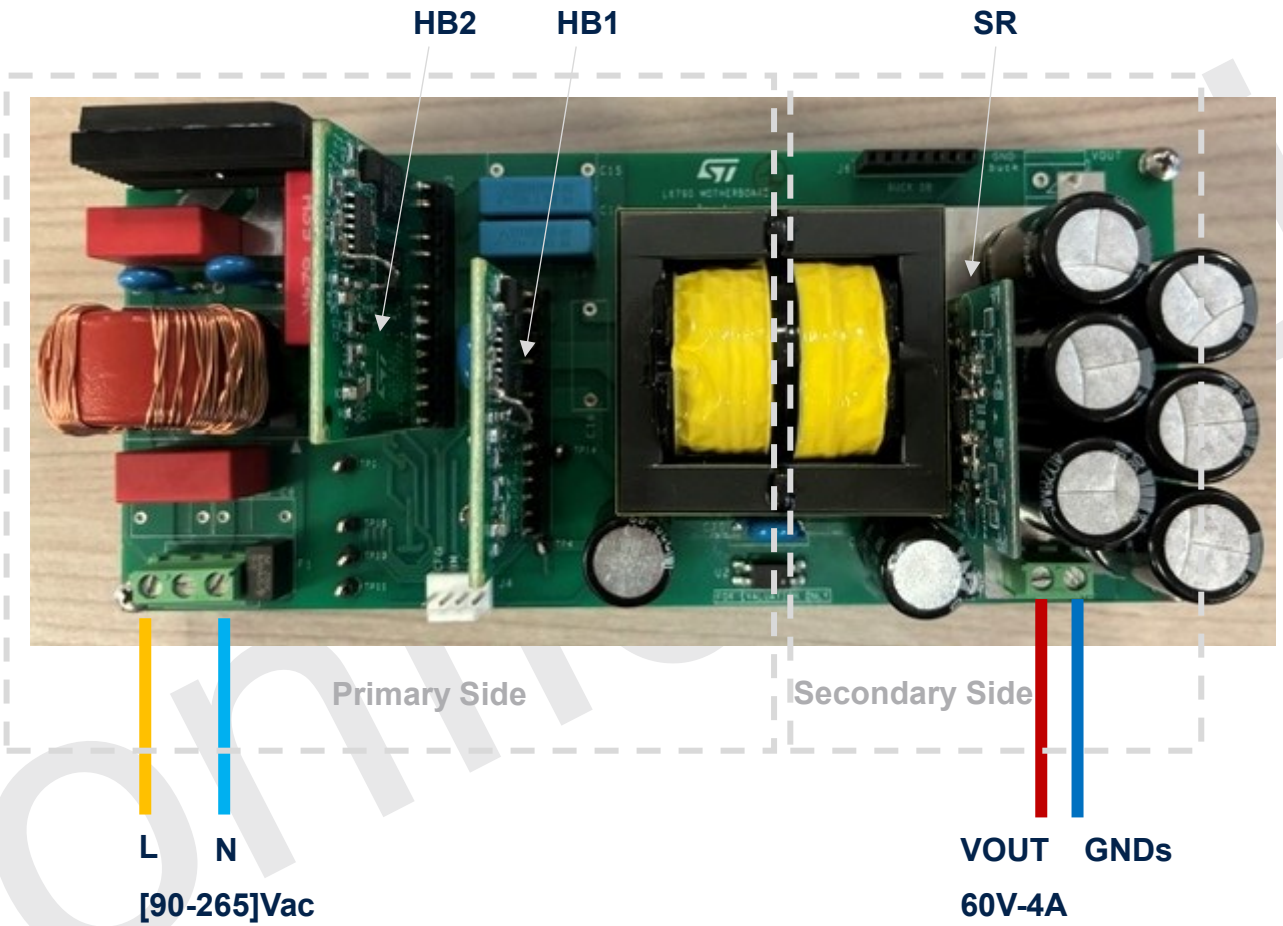
High Power Factor LLC evaluation board Rev.1

Applications Team

01/10/2026

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L6790 240W-60V-Evaluation board Rev.1



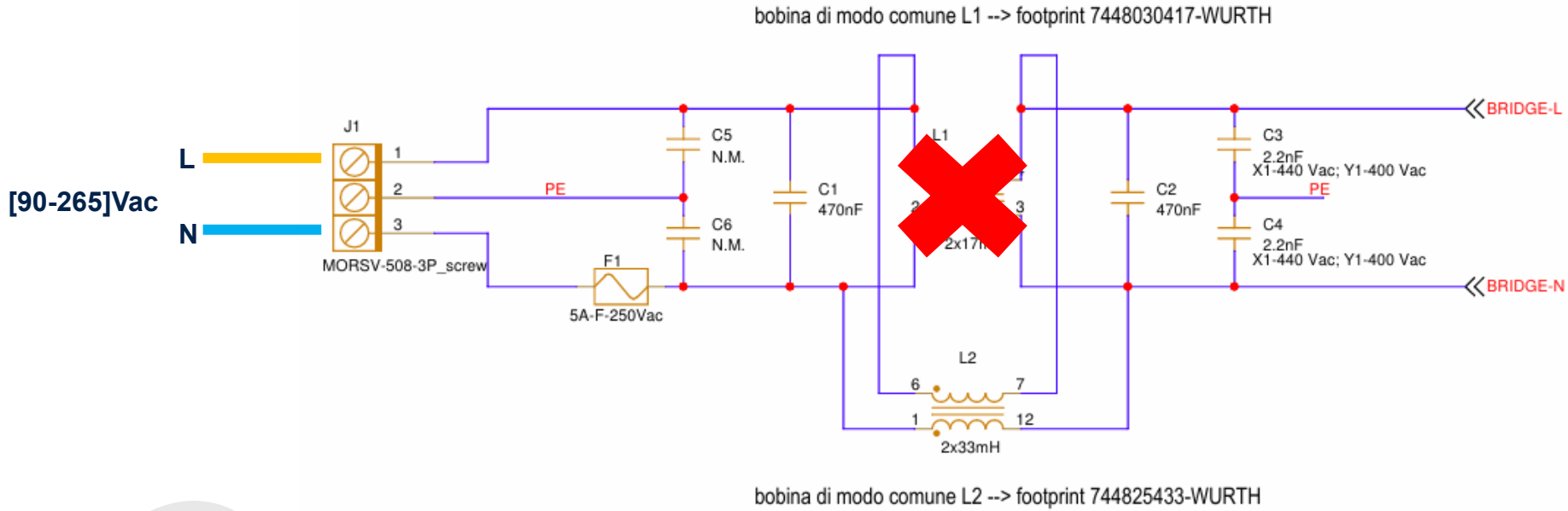
Note: for functional evaluation only

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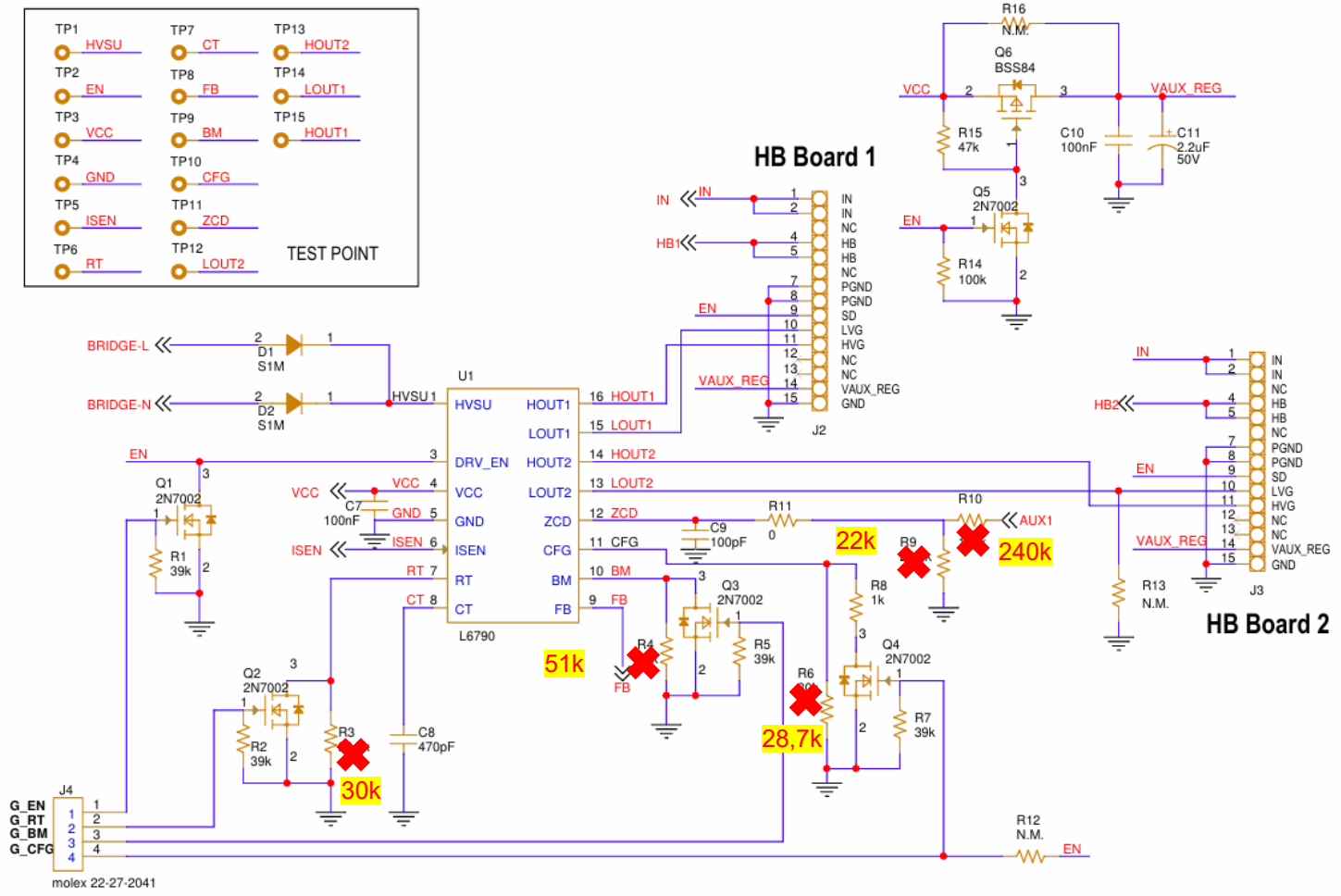
Schematics 1- EMI FILTER



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Schematics 2 - L6790_BOARD

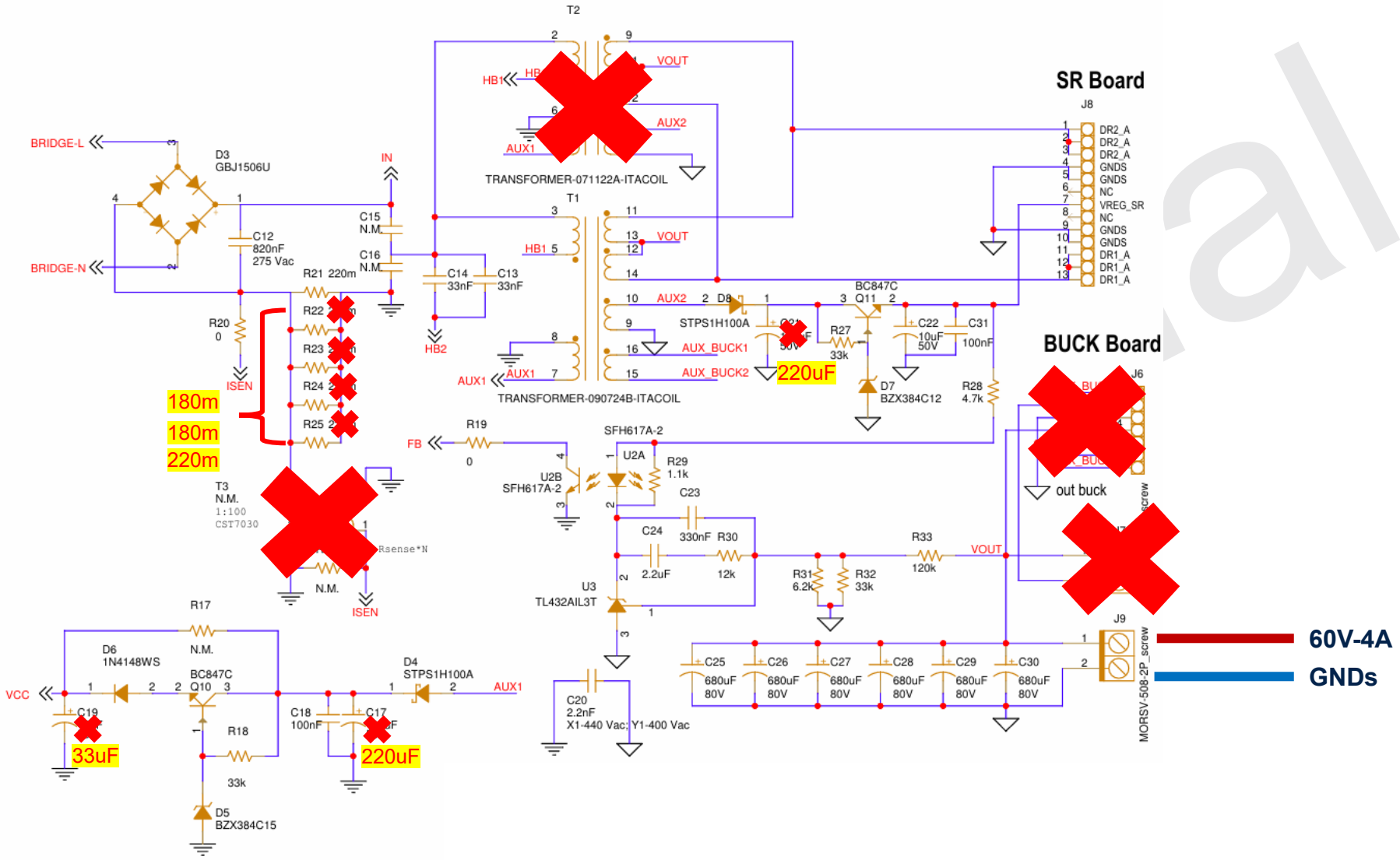


Note: L6790 ES vers. AA

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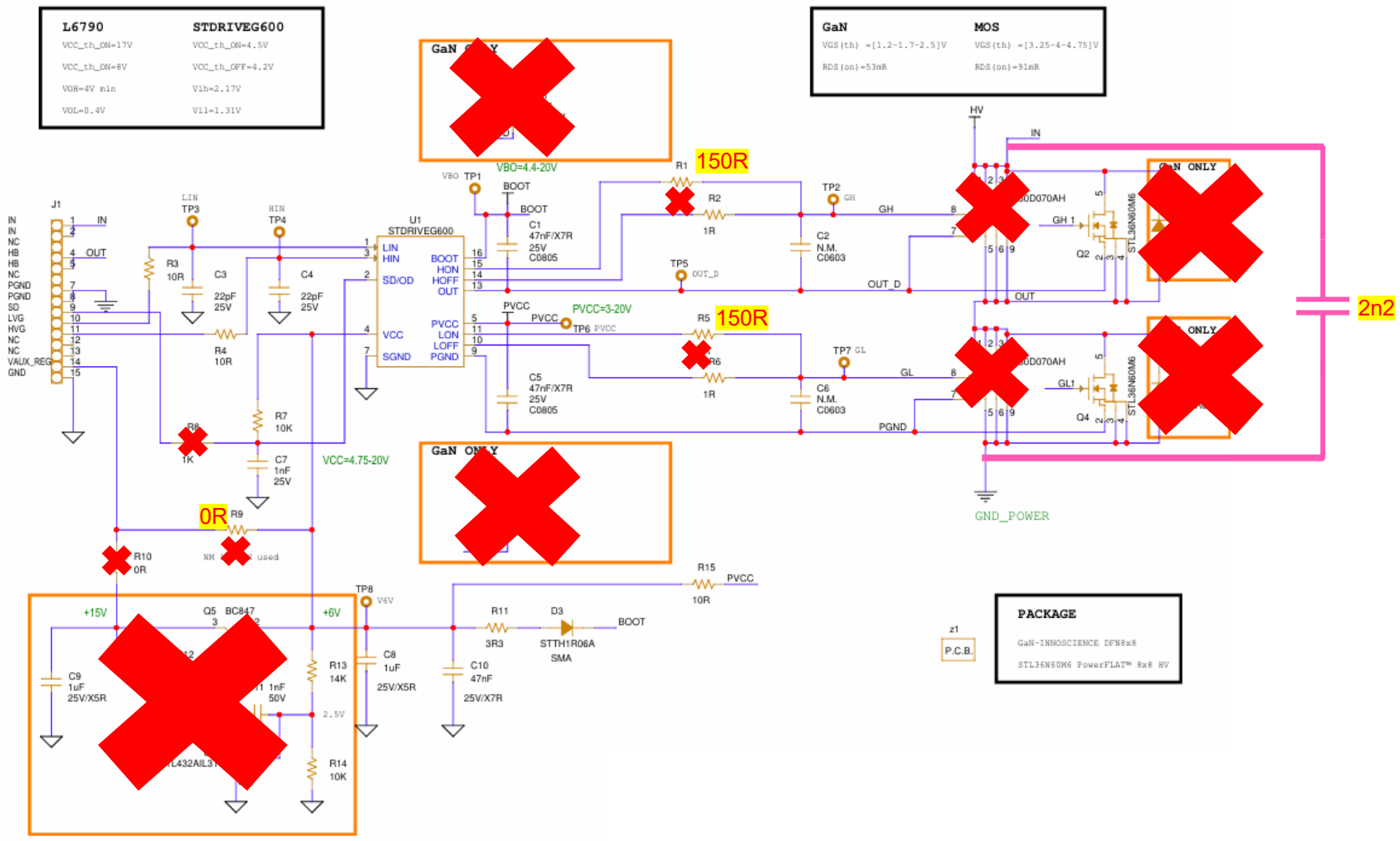
Schematics 3 - L6790_BOARD



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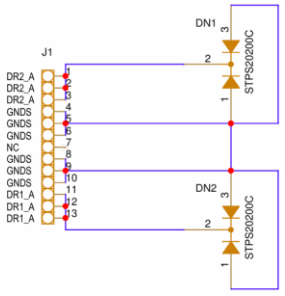
Schematics 4 – MOS-Half Bridge board



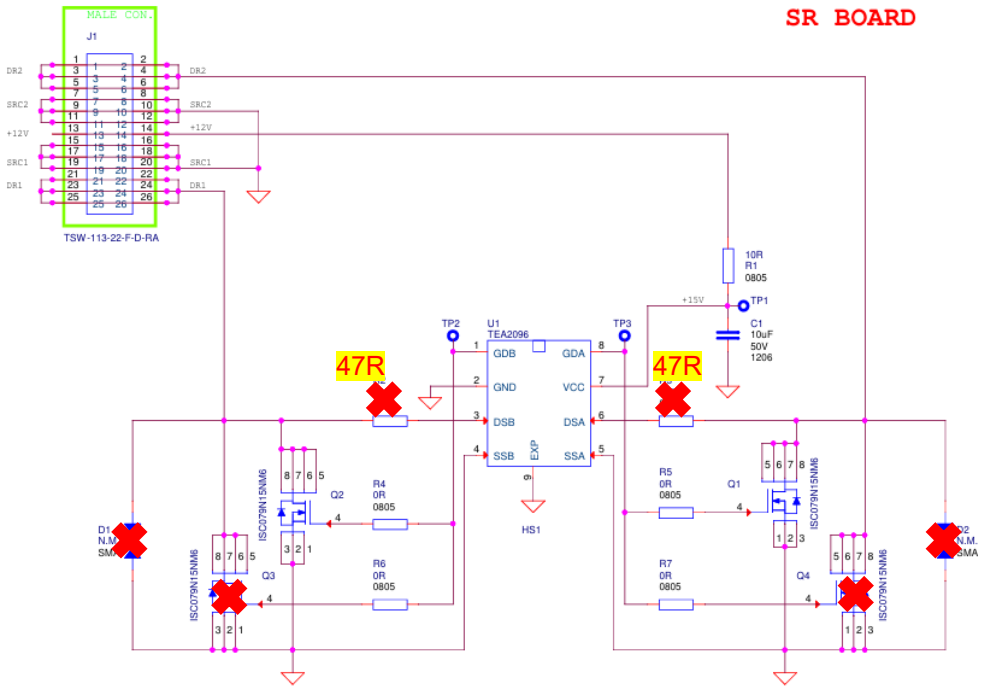
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Schematics 5 – SR board



*Alternative board for rectification



Note: SR board used for efficiency measurements

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ITACOIL Trafo Specs.

itacoilweb	DATASHEET	03/09/2026	REV	01	
	EDITED	Davide Maida	APPROVED	Dario Radaelli	

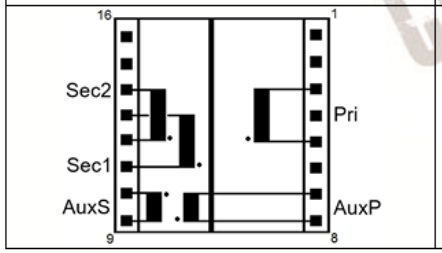
FINAL P/N	TBD	REV	00B	SAMPLING CODE	OP2207034-220726A
PRELIMINARY P/N	OP2207034			CUSTOMER P/N	
CUSTOMER	STMicroelectronics Srl				

PRELIMINARY * PRELIMINARY * PRELIMINARY

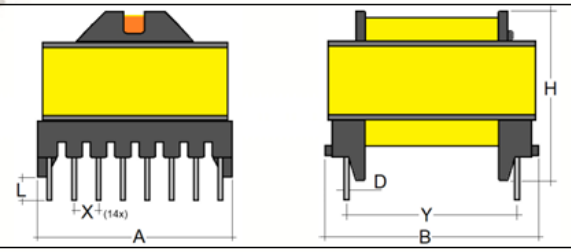
DESCRIPTION	ResPFC transformer

TEST/FEATURES			
Inductance Pri	225,0 µH ±10,0%	100KHz/1V	
Leakage Inductance Pri	52,2 µH ±10,0%	100KHz/1V	short-circuit on Sec1 or Sec2
Dielectric Strenght	Pri+AuxP / AuxS+Sec1+Sec2	3.8KV/2"	routine test
Dielectric Strenght	Pri+AuxP / AuxS+Sec1+Sec2	3,8KV/60"	type test
Turns ratio	Sec1:Pri:Sec2:AuxS:AuxP	1:2,9:1:0,4:0,4	

LAYOUT (bottom view)



DRAWING



DIMENSIONS (mm)									
A	43,5 max	B	47,2 max	H	36,5 max	X	5,0 typ	Y	35,0 typ
DØ	1,0 typ	L	3,0 min						

COMPONENT	THERMAL CLASS	CERTIFICATIONS
Materials compliant to UL electrical insulation system Itacoil 130-1	130°C (B)	E520270

NOTES

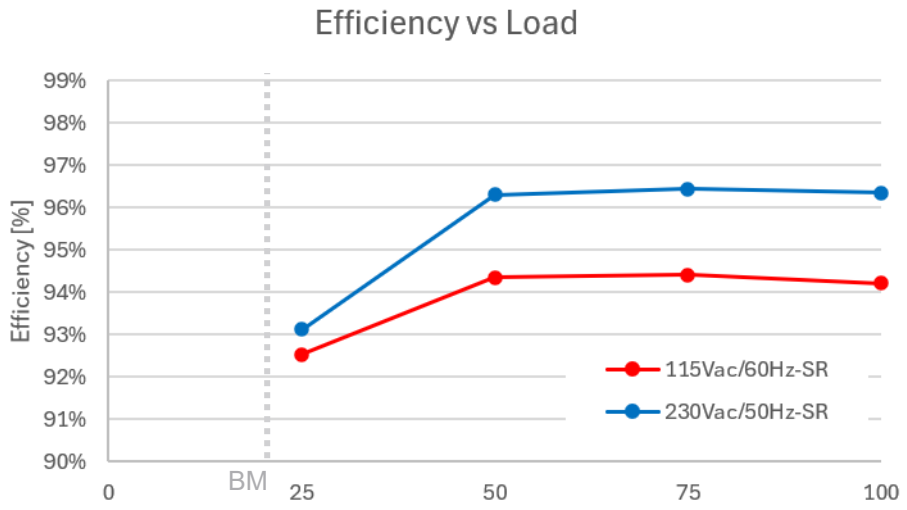
RoHS compliant [Directives 2011/65/EU and 2015/863/EU] see https://www.itacoilweb.com/files/RoHS.pdf
REACH compliant [Regulation (EC)1907/2006] see https://www.itacoilweb.com/regulatory-compliance/

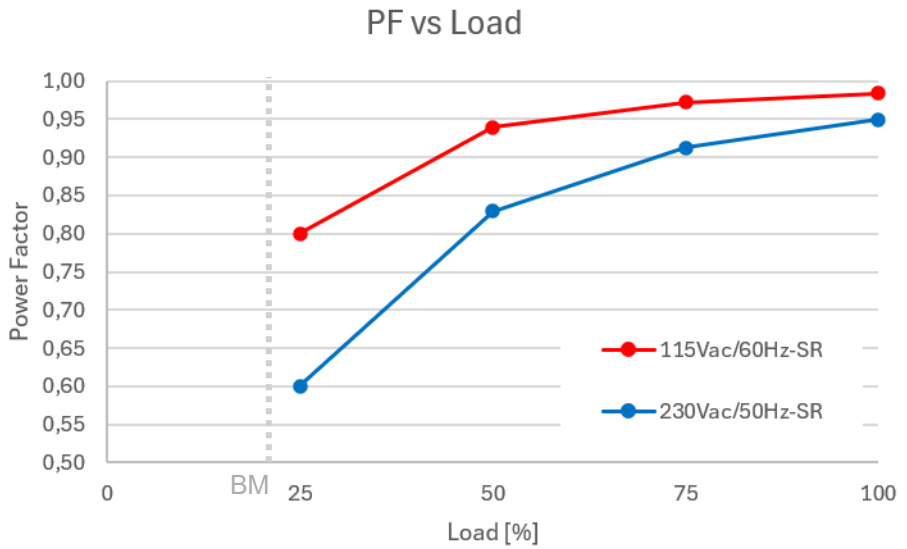
CONFIDENTIAL

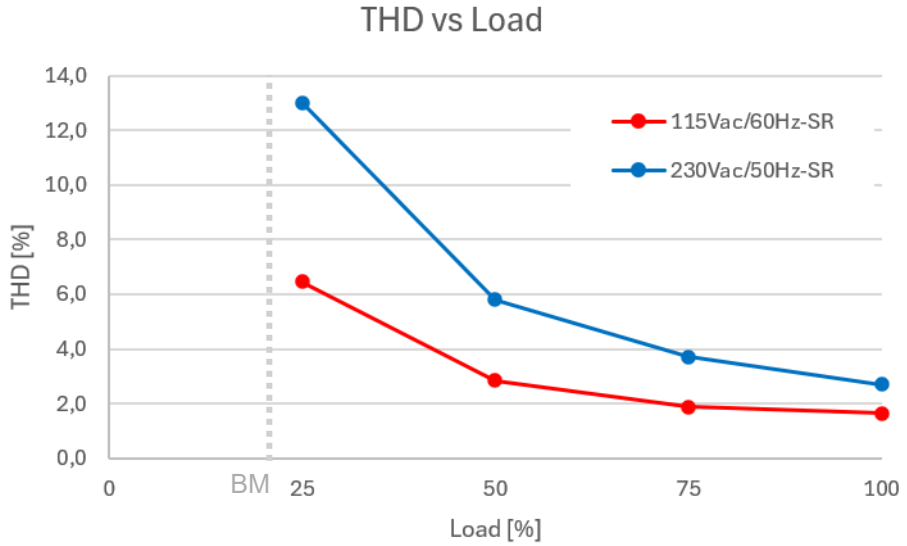
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Preliminary Measurements







$Pin_{_BM_th}$=56W

$P_{no\ load}$<180mW@230Vac

Note: optimizations on going

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